BRIGHTNESS AND CONTRAST GUIDED ADAPTIVE ENHANCEMENT FOR LOW-LIGHT IMAGES

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Abstract—Low-light images often suffer from poor visibility and reduced local contrast. This study implements a brightness- and contrast-guided adaptive enhancement (BCGAE) heuristic that selects gamma-correction and contrast-limited adaptive histogram equalization (CLAHE) parameters from image statistics. Four conventional methods—histogram equalization, gamma correction, CLAHE, and fixed gamma followed by CLAHE—are compared with BCGAE. Ten source images are synthetically darkened at mild, moderate, and severe levels, yielding 30 low-light inputs and 150 image-method evaluations. Mean squared error (MSE), peak signal-to-noise ratio (PSNR), structural similarity (SSIM), entropy, contrast, and processing time are recorded. Results show that no method dominates all metrics: histogram equalization achieves high PSNR in the tested synthetic setting but produces substantially higher contrast, while fixed gamma plus CLAHE gives strong SSIM at mild and moderate levels. BCGAE is competitive under severe degradation but incurs higher processing time than the single-operation baselines. The results illustrate the trade-off between visibility enhancement and similarity to a clean reference.

Keywords—Low-light image enhancement, gamma correction, histogram equalization, CLAHE, adaptive enhancement, image quality metrics.

I. INTRODUCTION

Images acquired under insufficient illumination may have low luminance, compressed intensity ranges, reduced local contrast, and visible loss of detail. Classical digital image processing offers low-cost enhancement techniques such as intensity transformations and histogram equalization. However, a method that improves visibility can also exaggerate contrast, alter colors, or diverge from the original reference. Surveys of low-light enhancement describe histogram-based, gray-level transformation, Retinex, fusion, and learning-based approaches, with no single method universally suitable for every image [3], [4].

This project studies a compact, interpretable alternative to fixed parameter selection. BCGAE measures input brightness and contrast and uses threshold-based rules to select a gamma value and CLAHE clip limit. Its contribution is an evaluated parameter-selection heuristic built from established operators, not a claim to introduce a new fundamental enhancement operator.

II. RELATED WORK

Pizer et al. studied adaptive histogram equalization and its variations, noting both the utility of local contrast enhancement and the risk of over-enhancing noise in homogeneous regions [1]. Zuiderveld described contrast-limited adaptive histogram equalization, which limits local histogram amplification to mitigate excessive contrast enhancement [2]. More recent reviews organize low-light enhancement methods into conventional and learning-based families and discuss the importance of both visual assessment and quantitative evaluation [3], [4].

Gamma correction applies a nonlinear intensity mapping and is computationally inexpensive, while global histogram equalization redistributes intensity values across the image. CLAHE operates on local regions and limits amplification. A fixed gamma-plus-CLAHE pipeline combines global brightness adjustment with local contrast enhancement, but fixed parameters may not suit every input. BCGAE addresses this parameter-selection issue through image-statistic-based rules.

III. PROPOSED METHODOLOGY

Let I be the input image. Brightness B is computed from its luminance representation, and contrast C is represented by the standard deviation of luminance. The current implementation chooses gamma using brightness thresholds: gamma = 0.45 if $B < 60$, 0.60 if $60 \leq B < 100$, and 0.75 otherwise. It selects CLAHE clip limit as 3.0 if $C < 35$, 2.0 if $35 \leq C < 60$, and 1.5 otherwise. These thresholds are heuristic settings used in this experiment; they are not claimed to be globally optimal.

First, gamma correction is applied using a 256-entry lookup table. Next, the image is converted to CIELAB, CLAHE is applied to the L channel, and the image is converted back to BGR. This luminance-channel operation aims to enhance local contrast while reducing direct manipulation of chromatic channels. The baseline methods use their corresponding conventional operations and the parameter settings in the experiment code.

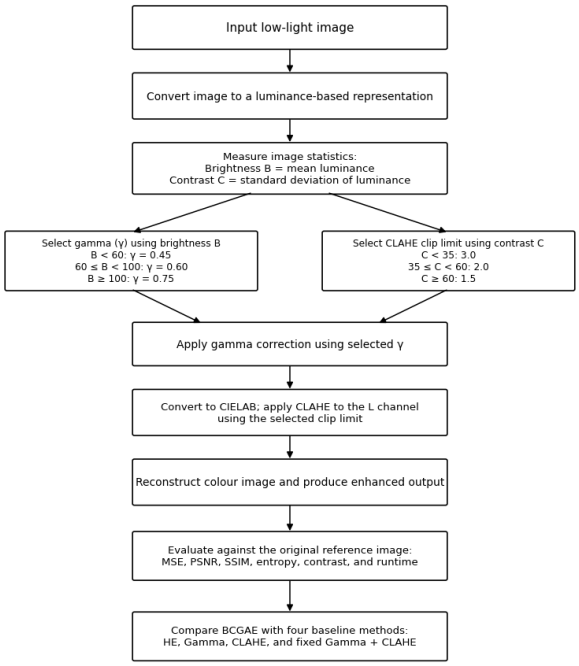


Fig. 1. Flowchart of the proposed Brightness and Contrast Guided Adaptive Enhancement (BCGAE) methodology.

IV. EXPERIMENTAL SETUP

The experiment uses ten source images and synthetically darkens each source at three severity levels (mild, moderate, severe). The source image is retained as the reference. Five enhancement methods are evaluated at each level, resulting in $10 \times 3 \times 5 = 150$ image-method records. The reported averages are calculated from the supplied evaluation CSV. This controlled setup supports

reference-based metrics, but does not fully reproduce sensor noise, white-balance shifts, motion blur, or other effects of real low-light photography.

MSE measures squared pixel error (lower is better); PSNR is derived from MSE (higher is better); SSIM compares structural similarity (higher is generally better). Entropy summarizes intensity-distribution information and contrast describes luminance spread; neither should be interpreted as universally better when maximized. Runtime is measured in milliseconds and depends on hardware, image dimensions, and software environment.

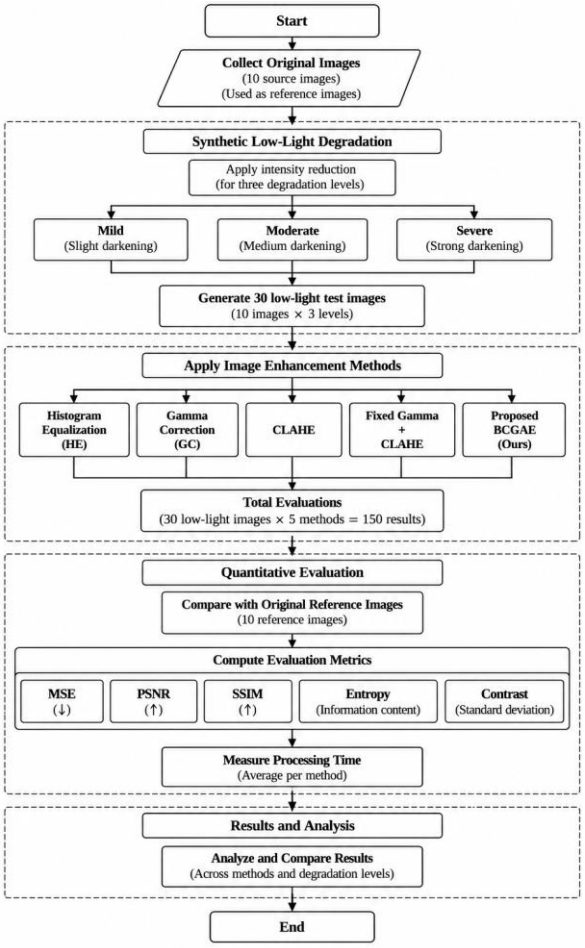


Fig. 2. Experimental workflow illustrating dataset preparation, synthetic low-light degradation at three severity levels, application of five image enhancement methods, quantitative evaluation using MSE, PSNR, SSIM, entropy, and contrast, and comparative analysis of the results.

V. RESULTS AND DISCUSSION

Table I summarizes the mean MSE, PSNR, and SSIM for each degradation level. The results show that performance depends on both degradation severity and metric. Under mild degradation, fixed gamma plus CLAHE has the lowest MSE (994.35) achieves the highest PSNR (18.8342 dB), narrowly exceeding gamma correction (18.8263 dB), while CLAHE records the highest SSIM (0.9121). BCGAE does not lead the mild-level reference metrics.

At the moderate level, histogram equalization records the lowest MSE (1330.85) and highest PSNR (18.13 dB), while fixed gamma plus CLAHE obtains the highest SSIM (0.8946). At severe degradation, BCGAE records the lowest MSE (1473.52) and highest SSIM (0.8156) among the tested methods, with histogram equalization obtaining a slightly higher PSNR (17.30 dB versus 17.06 dB). Histogram equalization also produces much higher contrast (70.91) than BCGAE (33.57), which may indicate aggressive tonal expansion rather than universally superior visual quality. The figures should therefore be interpreted together with visual comparisons.

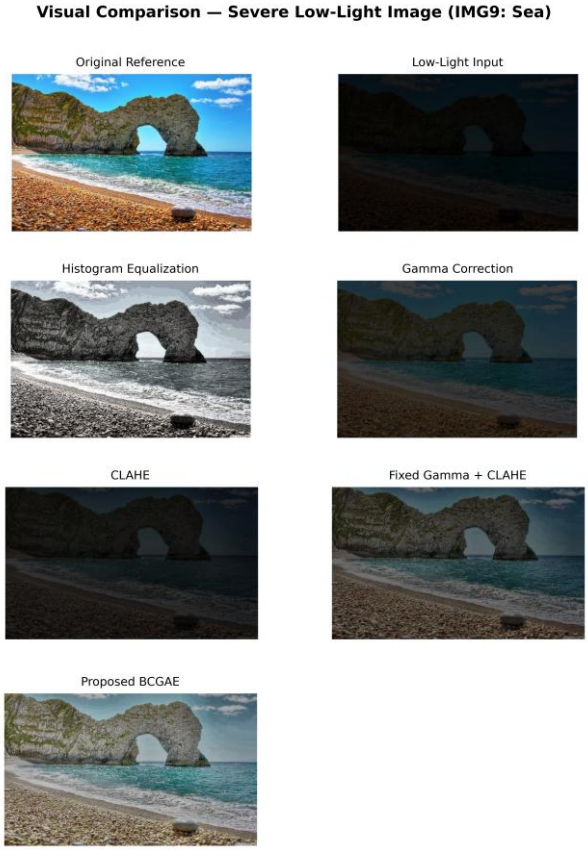


Fig. 3. Visual comparison of five enhancement methods on a severely degraded low-light image (IMG9—Sea).

Level	Method	MSE ↓	PSNR (dB) ↑	SSIM ↑
Mild	CLAHE	1865.79	16.53	0.9121
Mild	Gamma + CLAHE	994.35	18.83	0.8900
Mild	Gamma	985.06	18.83	0.9028
Mild	HE	1246.09	18.31	0.8402
Mild	BCGAE	1917.49	16.04	0.8319
Moderate	CLAHE	5830.87	10.92	0.6556
Moderate	Gamma + CLAHE	2083.14	15.91	0.8946
Moderate	Gamma	3903.42	12.65	0.7387
Moderate	HE	1330.85	18.13	0.8192
Moderate	BCGAE	1873.40	15.83	0.8261
Severe	CLAHE	11137.45	7.97	0.2792
Severe	Gamma + CLAHE	5332.95	11.47	0.7088
Severe	Gamma	8184.60	9.33	0.4594
Severe	HE	1704.58	17.30	0.7362
Severe	BCGAE	1473.52	17.06	0.8156

TABLE I. MEAN REFERENCE-BASED METRICS BY DEGRADATION LEVEL

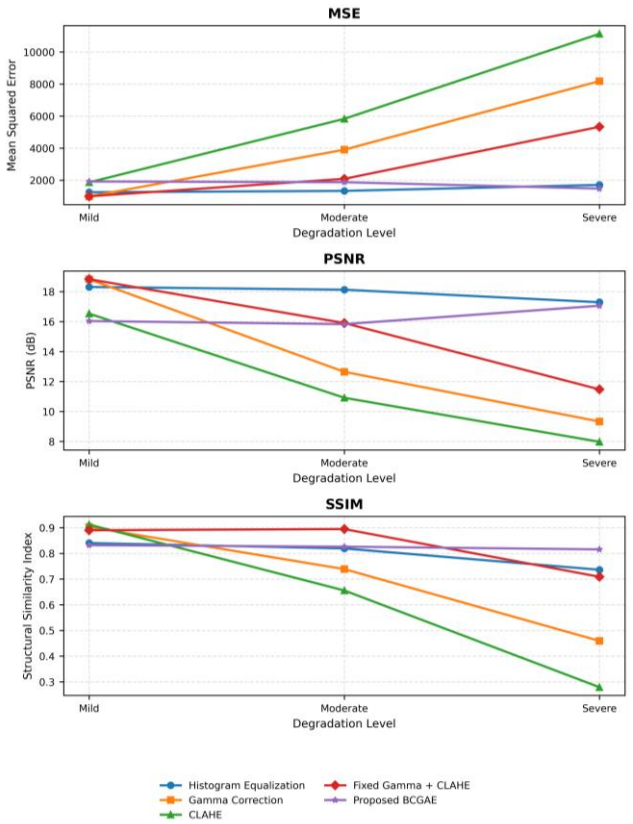


Fig. 4. Quantitative comparison of five low-light image enhancement methods across mild, moderate, and severe degradation levels: (a) MSE, (b) PSNR, and (c) SSIM. Values represent the mean across ten test images per degradation level.

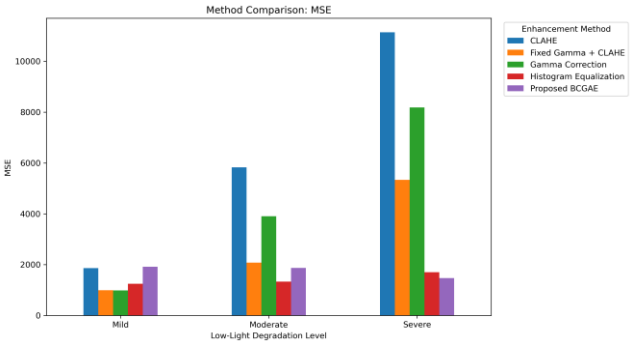


Fig. 5. Comparison of mean squared error (MSE) for five image enhancement methods across mild, moderate, and severe low-light degradation levels. Lower MSE indicates smaller pixel-wise differences from the original reference images.

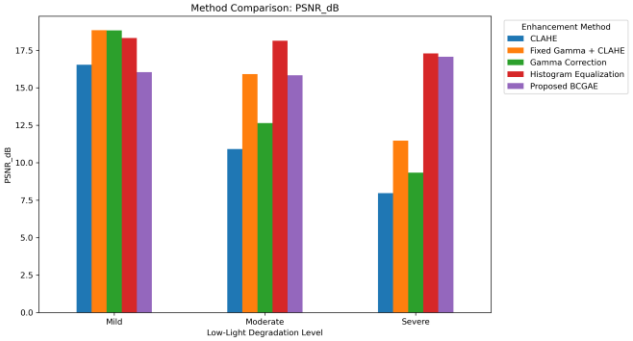


Fig. 6. Comparison of peak signal-to-noise ratio (PSNR) for five image enhancement methods across mild, moderate, and severe low-light degradation levels. Higher PSNR indicates greater pixel-level fidelity to the original reference images.

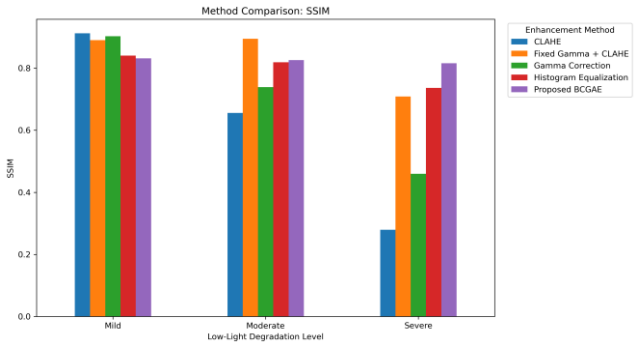


Fig. 7. Comparison of structural similarity index (SSIM) for five image enhancement methods across mild, moderate, and severe low-light degradation levels. Higher SSIM indicates greater structural similarity to the original reference images.

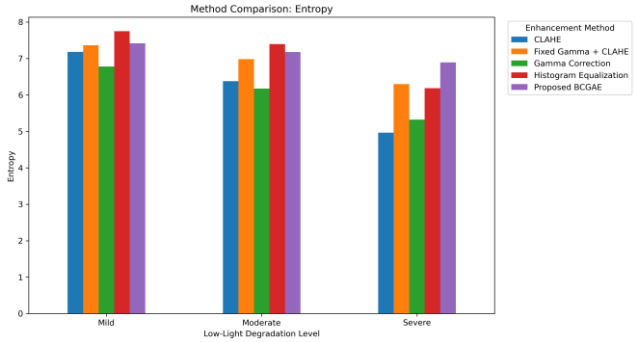


Fig. 8. Comparison of image entropy for five enhancement methods across mild, moderate, and severe low-light degradation levels. Entropy characterizes the distribution of intensity information; higher entropy does not necessarily indicate better perceptual quality.

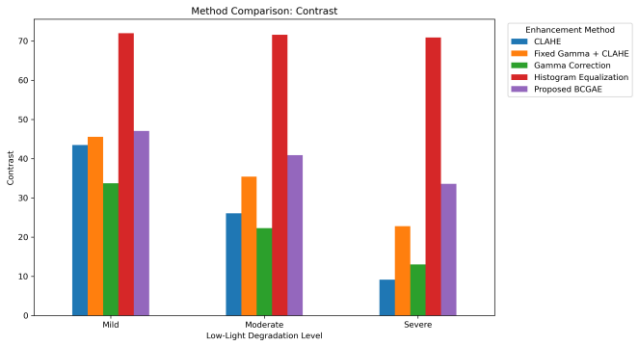


Fig. 9. Comparison of luminance contrast, measured using standard deviation, for five enhancement methods across mild, moderate, and severe low-light degradation levels. Contrast values should be interpreted alongside other image quality metrics.

The performance of the five image enhancement methods was evaluated across mild, moderate, and severe low-light degradation levels. Visual comparisons were performed to examine enhancement quality, while quantitative evaluation was conducted using MSE, PSNR, SSIM, entropy, and contrast. Processing time was also measured to assess computational efficiency. The combined and individual metric plots are presented to facilitate comparison across methods and degradation levels.

Overall, the results indicate that enhancement performance varies with degradation severity and evaluation metric, with BCGAE performing competitively under severe degradation but not consistently outperforming all baseline methods.

A. Processing Time

The timing summary reports mean processing times of 0.544 ms for gamma correction, 3.167 ms for histogram equalization, 3.311 ms for CLAHE, 3.893 ms for fixed gamma plus CLAHE, and 6.770 ms for BCGAE. BCGAE is the slowest among the evaluated methods in this timing run, consistent with the extra brightness/contrast analysis and parameter-selection steps. Runtime comparisons should be considered environment-specific.

Method	Mean (ms)	Median (ms)
Gamma Correction	0.544	0.308
Histogram Equalization	3.167	1.788
CLAHE	3.311	1.835
Fixed Gamma + CLAHE	3.893	2.234
Proposed BCGAE	6.770	3.766

TABLE II. MEASURED PROCESSING TIME

B. Limitations

The test images are synthetically darkened, so the experiment does not establish performance on real low-light captures. The heuristic thresholds were selected manually and have not been optimized across a separate validation set. Reference metrics may penalize visually useful changes, while high entropy or contrast may accompany artifacts. Future work should include real low-light datasets, more images, parameter sensitivity analysis, and a blinded visual assessment.

VI. CONCLUSION

A brightness- and contrast-guided enhancement pipeline combining gamma correction and CLAHE was implemented and compared with four conventional alternatives. The measured results demonstrate that method rankings change with degradation level and evaluation metric. BCGAE is a simple, interpretable heuristic rather than a universally superior method; its main value is adaptive parameter selection and a reproducible comparative evaluation. Broader validation on real low-light images is required before drawing general conclusions.

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